1. Remove First Element

```
ArrayList<String> cities = new ArrayList<>();

cities.add("Chennai");
cities.add("Salem");
cities.add("Coimbatore");

cities.remove(0);

System.out.println(cities);
```

2. Check City Exists

```
ArrayList<String> cities = new ArrayList<>();

cities.add("Chennai");
cities.add("Salem");
cities.add("Madurai");

System.out.println(cities.contains("Salem"));
```

3. Print All Student Names

```
ArrayList<String> names = new ArrayList<>();

names.add("Saran");
names.add("Kumar");
names.add("Priya");

for(int i = 0; i < names.size(); i++) {
   System.out.println(names.get(i));
}
```

19. Sum of 3 Numbers

```
ArrayList<Integer> nums = new ArrayList<>();

nums.add(10);
```

```java
nums.add(20);
nums.add(30);

int sum = 0;

for(int i = 0; i < nums.size(); i++) {
    sum += nums.get(i);
}

System.out.println(sum);
```

Output

60

20. Find Largest Number

```java
ArrayList<Integer> nums = new ArrayList<>();

nums.add(15);
nums.add(50);
nums.add(25);

int max = nums.get(0);

for(int i = 1; i < nums.size(); i++) {
    if(nums.get(i) > max) {
        max = nums.get(i);
    }
}

System.out.println(max);
```

21. Print Even Numbers

```java
ArrayList<Integer> nums = new ArrayList<>();

nums.add(10);
nums.add(15);
nums.add(20);
```

```
nums.add(25);

for(int i = 0; i < nums.size(); i++) {
    if(nums.get(i) % 2 == 0) {
        System.out.println(nums.get(i));
    }
}
```

22. Count Elements Greater Than 50

```
ArrayList<Integer> nums = new ArrayList<>();

nums.add(20);
nums.add(80);
nums.add(60);
nums.add(10);

int count = 0;

for(int i = 0; i < nums.size(); i++) {
    if(nums.get(i) > 50) {
        count++;
    }
}

System.out.println(count);
```

23. Reverse Print

```
ArrayList<String> names = new ArrayList<>();

names.add("A");
names.add("B");
names.add("C");

for(int i = names.size()-1; i >= 0; i--) {
    System.out.println(names.get(i));
}
```

24. Find First and Last Element

```
ArrayList<String> fruits = new ArrayList<>();

fruits.add("Apple");
fruits.add("Mango");
fruits.add("Banana");

System.out.println("First: " + fruits.get(0));
System.out.println("Last: " + fruits.get(fruits.size()-1));
```

25. Clear All Elements

```
ArrayList<String> names = new ArrayList<>();

names.add("Saran");
names.add("Kumar");

names.clear();

System.out.println(names);
```

Output

```
[]
```

Secret Agent Mission

Question

3 secret codes add பண்ணுங்க.

```
import java.util.ArrayList;

import java.util.Scanner;


public class Main {

    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
        ArrayList<String> secretCodes = new ArrayList<>();

        System.out.println("Enter 3 Secret Codes:");

        for(int i = 0; i < 3; i++) {

            String code = sc.nextLine();
            String encrypted = "";

            for(int j = 0; j < code.length(); j++) {
                char ch = code.charAt(j);
                encrypted += (char)(ch + 1);
            }

            secretCodes.add(encrypted);
        }

        System.out.println("\nEncrypted Codes Stored:");

        for(String code : secretCodes) {
            System.out.println(code);
        }
    }
}
```

Sample Input

AGENT

HELLO
1234

Stored in ArrayList

[BHFOU, IFMMP, 2345]

Output

Encrypted Codes Stored:
BHFOU
IFMMP
2345

Food Ordering App

Question

Customer order list create பண்ணுங்க.

```
import java.util.ArrayList;

import java.util.Scanner;


public class Main {

    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);

        ArrayList<String> orders = new ArrayList<>();


        System.out.println("Enter 5 Food Orders:");


        for(int i = 0; i < 5; i++) {

            String food = sc.nextLine();

            orders.add(food);

        }
```

```
        System.out.println("\nCustomer Orders:");

        for(String order : orders) {
            System.out.println(order);
        }
    }
}
```

Alphabet A Star Pattern:

```
public class APattern {
    public static void main(String[] args) {

        int n = 5;

        // Upper Part
        for (int i = 0; i < n; i++) {

            // Spaces
            for (int j = 0; j < n - i - 1; j++) {
                System.out.print(" ");
            }

            // Stars
            for (int j = 0; j < 2 * i + 1; j++) {

                if (j == 0 || j == 2 * i ||
```

```java
                (i == n / 2 && j > 0 && j < 2 * i)) {
                System.out.print("*");
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }

    // Lower Part
    for (int i = 0; i < n - 1; i++) {

        System.out.print(" ");

        for (int j = 0; j < 2 * n - 1; j++) {

            if (j == 0 || j == 2 * n - 2) {
                System.out.print("*");
            } else {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
  }
}
```

Alphabet B Star Pattern:

```
public class BPattern {
    public static void main(String[] args) {

        int rows = 7;
        int cols = 6;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (i == 0 || i == rows / 2 || i == rows - 1) {
                    System.out.print("*");
                }
                else if (j == 0 || j == cols - 1) {
                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
            }

            System.out.println();
        }
    }
}
```

Alphabet C Star Pattern:

```
public class CPattern {
    public static void main(String[] args) {

        int rows = 7;
        int cols = 7;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (i == 0 || i == rows - 1) {
                    System.out.print("*");
                }
                else if (j == 0) {
                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
            }

            System.out.println();
        }
    }
}
```

Alphabet C Star Pattern:

```
public class DPattern {
    public static void main(String[] args) {

        int rows = 7;
        int cols = 7;

        for (int i = 0; i < rows; i++) {

            for (int j = 0; j < cols; j++) {

                if (j == 0) { // Left vertical line
                    System.out.print("*");
                }
                else if ((i == 0 || i == rows - 1) && j < cols - 1) {
                    System.out.print("*");
                }
                else if (j == cols - 1 &&
                        i != 0 &&
                        i != rows - 1) {
                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
```

```
            }

            System.out.println();
        }
    }
}
```